

Problem 1

Part (a)

i) If B attracts A , then A also attracts B :

Let B attract A , i.e., $\mathbb{P}(A|B) > \mathbb{P}(A)$ and $\mathbb{P}(B) > 0$.

First, we show that $\mathbb{P}(A) > 0$. Assume for contradiction that $\mathbb{P}(A) = 0$. Since $A \cap B \subseteq A$, then $\mathbb{P}(A \cap B) \leq \mathbb{P}(A) = 0$ and also $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = 0$. This contradicts the assumption that $\mathbb{P}(A|B) > \mathbb{P}(A) = 0$. Therefore, $\mathbb{P}(A) > 0$.

Second, we show that A attracts B :

$$\begin{aligned} & \mathbb{P}(A|B) > \mathbb{P}(A) \\ \iff & \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} > \mathbb{P}(A) \\ \iff & \mathbb{P}(A \cap B) > \mathbb{P}(A)\mathbb{P}(B) \\ \iff & \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} > \mathbb{P}(B) \quad (\text{since } \mathbb{P}(A) > 0) \\ \iff & \mathbb{P}(B|A) > \mathbb{P}(B) \end{aligned}$$

□

ii) If B attracts A , then B^c repels A :

Let B attract A , i.e., $\mathbb{P}(A|B) > \mathbb{P}(A)$, which implies $\mathbb{P}(B) > 0$.

First, we show that $\mathbb{P}(B^c) > 0$. Assume for contradiction that $\mathbb{P}(B^c) = 0$. Then $\mathbb{P}(B) = 1$. Since $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$ and $\mathbb{P}(A \cap B^c) \leq \mathbb{P}(B^c) = 0$, this implies $\mathbb{P}(A \cap B) = \mathbb{P}(A)$. Then $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A)$, contradicting $\mathbb{P}(A|B) > \mathbb{P}(A)$. Therefore, $\mathbb{P}(B^c) > 0$.

Second, we show that B^c repels A :

$$\begin{aligned} & \mathbb{P}(A|B) > \mathbb{P}(A) \\ \implies & \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} > \mathbb{P}(A) \\ \implies & \mathbb{P}(A \cap B) > \mathbb{P}(A)\mathbb{P}(B) \end{aligned}$$

Since B and B^c form a partition of Ω , then $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$. Substituting $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) + \mathbb{P}(A \cap B^c)$ yields:

$$\begin{aligned} & \mathbb{P}(A) - \mathbb{P}(A \cap B^c) > \mathbb{P}(A)\mathbb{P}(B) \\ \implies & \mathbb{P}(A \cap B^c) < \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) \\ \implies & \mathbb{P}(A \cap B^c) < \mathbb{P}(A)(1 - \mathbb{P}(B)) \\ \implies & \mathbb{P}(A \cap B^c) < \mathbb{P}(A)\mathbb{P}(B^c) \\ \implies & \frac{\mathbb{P}(A \cap B^c)}{\mathbb{P}(B^c)} < \mathbb{P}(A) \quad (\text{since } \mathbb{P}(B^c) > 0) \\ \implies & \mathbb{P}(A|B^c) < \mathbb{P}(A) \end{aligned}$$

□

iii) If A attracts B and B attracts C , does A then also attract C ?

No. We prove this by counterexample.

Let the probability space be $(\Omega, \mathcal{F}, \mathbb{P})$ with $\Omega = \{1, 2, 3\}$, $\mathcal{F} = \mathcal{P}(\Omega)$ and $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$, $\forall A \in \mathcal{F}$ (Laplace probability space). Define the events:

$$\begin{aligned} A = \{1, 2\} &\implies \mathbb{P}(A) = \frac{2}{3} \\ B = \{2\} &\implies \mathbb{P}(B) = \frac{1}{3} \\ C = \{2, 3\} &\implies \mathbb{P}(C) = \frac{2}{3} \end{aligned}$$

We verify that A attracts B and B attracts C , but A repels C :

$$\begin{aligned} \mathbb{P}(A \cap B) &= \frac{1}{3} > \frac{2}{9} = \mathbb{P}(A)\mathbb{P}(B) \\ \implies \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} &> \mathbb{P}(A) \\ \implies \mathbb{P}(A|B) &> \mathbb{P}(A) && (A \text{ attracts } B) \\ \mathbb{P}(B \cap C) &= \frac{1}{3} > \frac{2}{9} = \mathbb{P}(B)\mathbb{P}(C) \\ \implies \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(C)} &> \mathbb{P}(B) \\ \implies \mathbb{P}(B|C) &> \mathbb{P}(B) && (B \text{ attracts } C) \\ \mathbb{P}(A \cap C) &= \frac{1}{3} < \frac{4}{9} = \mathbb{P}(A)\mathbb{P}(C) \\ \implies \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(C)} &< \mathbb{P}(A) \\ \implies \mathbb{P}(A|C) &< \mathbb{P}(A) && (A \text{ repels } C) \end{aligned}$$

Therefore, in general, if A attracts B and B attracts C , it does not necessarily follow that A attracts C . □

b) Laplace experiment with prime cardinality:

Let the probability space be a Laplace experiment $(\Omega, \mathcal{P}(\Omega), \mathbb{P})$ where $\Omega = \{1, 2, \dots, p\}$, $|\Omega| = p$ (with p being a prime number), and $\mathbb{P}(E) = \frac{|E|}{p}$ for any $E \in \mathcal{P}(\Omega)$.

Assume $A, B \in \mathcal{P}(\Omega)$ are independent. Then:

$$\begin{aligned} \mathbb{P}(A \cap B) &= \mathbb{P}(A)\mathbb{P}(B) \\ \implies \frac{|A \cap B|}{p} &= \frac{|A|}{p} \cdot \frac{|B|}{p} \\ \implies |A \cap B| &= \frac{|A| \cdot |B|}{p} \end{aligned}$$

Because $|A \cap B| = \frac{|A| \cdot |B|}{p} \in \mathbb{N}_0$, $|A| \cdot |B|$ must be divisible by p . Since p is a prime number, the only way in which p , the denominator, can be cancelled is if one of the factors, $|A|$ or $|B|$, is divisible by p .

Since $A, B \subseteq \Omega$, then $0 \leq |A|, |B| \leq p$. Therefore, $|A|, |B| \in \{0, 1, \dots, p\}$. The only numbers in $\{0, 1, \dots, p\}$ divisible by p are 0 and p . Therefore:

$$|A| \in \{0, p\} \quad \text{or} \quad |B| \in \{0, p\}$$

If the cardinality is 0, the event is the empty set $\emptyset$. If the cardinality is p , the event is the entire sample space Ω . Therefore, at least one of the events A or B is $\emptyset$ or Ω .

□

c) Upper bound for the non-occurrence of independent events:

The event that none of the A_i occurs is given by the intersection of their complements: $\bigcap_{i=1}^n A_i^c$.

First, we show that the complements are independent. Let $\mathcal{M}_i = \{A_i\}$ for $i = 1, \dots, n$. Since $A_1, \dots, A_n$ are independent, the family of set systems $(\mathcal{M}_i)_{i=1}^n$ is independent (Definition 2.2.9). Furthermore, each $\mathcal{M}_i$ is intersection-stable because it contains only a single element. Because independent, intersection-stable set systems generate independent σ -algebras (Lemma 2.2.11), the family of generated σ -algebras $(\sigma(\mathcal{M}_i))_{i=1}^n$, with $\sigma(\mathcal{M}_i) = \{\emptyset, A_i, A_i^c, \Omega\}$, is independent. Since $A_i^c \in \sigma(\mathcal{M}_i)$ for all $i = 1, \dots, n$, it follows that the events $A_1^c, \dots, A_n^c$ are independent (Definition 2.2.9).

Then:

$$\begin{aligned} \mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) &= \prod_{i=1}^n \mathbb{P}(A_i^c) = \prod_{i=1}^n (1 - \mathbb{P}(A_i)) \\ &\leq \prod_{i=1}^n \exp(-\mathbb{P}(A_i)) \quad (\text{since } 1 - x \leq \exp(-x) \text{ for } x = \mathbb{P}(A_i)) \\ &= \exp\left(-\sum_{i=1}^n \mathbb{P}(A_i)\right) \end{aligned}$$

Thus, the probability that none of the events $A_1, \dots, A_n$ occurs is less than or equal to $\exp(-\sum_{i=1}^n \mathbb{P}(A_i))$.

□